QUANTIFYING EXOPLANET HABITABILITY USING A COMPOSITE INDEX FRAMEWORK

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ABSTRACT

The discovery of exoplanets has significantly changed the search for life beyond Earth from speculation into supported science. With thousands of confirmed planets with a wide range of masses, sizes, orbital configurations and stellar environments, the question is no longer whether planets exist, but which of them may resemble Earth in meaningful ways. Habitability, however, cannot be observed directly. It must be inferred from measurable physical characteristics that influence the long-term presence of surface liquid water.

This study develops a composite habitability index constructed from planetary size, stellar energy input, orbital stability and host star properties using confirmed exoplanets from the NASA Exoplanet Archive. Each dimension is scaled between 0 and 1 and combined through a geometric mean to produce a single quantitative score. The distribution of this index is then examined across discovery years and detection methods to evaluate how potentially habitable planets have evolved over time. Using these methods and structural framework helps assess if the advancements in the technology are beneficial in detecting increased number of Earth-like habitable environments.

1 INTRODUCTION

The discovery of exoplanets has transformed astronomy over the past three decades. Since the first confirmed detection of a planet orbiting a Sun-like star in 1995, the number of known exoplanets has grown into the thousands. These planets show a wide range of diversity including mass, size, orbital distance and stellar environment. Although, these increase planets and their systems differ significantly from the environments on Earth.

Direct detection of life beyond Earth remains beyond current technological capability. As a result, planetary habitability is a concept that is indirectly inferred through observable physical characteristics that influence the long-term presence of surface liquid water on a planet. Planetary size and mass provide insight into composition, distinguishing rocky bodies from gas giants. Stellar energy input determines thermal conditions necessary for liquid water. Orbital eccentricity affects climate stability, while host star properties shape both habitable zone location and long-term environmental stability. Statistical analyses of confirmed exoplanets show that Earth-like structural conditions are rare within the detected bodies. Most known planets are either highly gas-dominated bodies or small rocky planets subjected to extreme stellar heating. Although some planets fall within conservative temperature ranging from compatible with liquid water, very few simultaneously satisfy structural, orbital and stellar criteria associated with Earth.

At the same time, observational capabilities have evolved with the changing modern technology. Early detection methods included large, closer planets. Recent missions on the other hand included planets such as Kepler and TESS that have improved sensitivity to smaller and cooler planets, expanding the scope in the outer space. These tcehnological advancements thus raises the following question:

How has the distribution of a quantitative habitability index for discovered exoplanets evolved with discovery year and across different detection facilities and methods?

To address this question, a composite habitability index is constructed using planetary and stellar variables. The index integrates size, stellar energy input, orbital stability and stellar environment into a scalar score between 0 and 1. Temporal analysis of this index across discovery years and detection methods allows evaluation of changes in both average habitability and the frequency of high-scoring planets. By combining multivariate index construction with time-series analysis, this study provides a quantitative framework for assessing how the statistical landscape of planetary habitability has evolved alongside advances in exoplanet detection.

2 METHODS

A composite index is defined using planetary and stellar parameters. Each confirmed exoplanet is assigned a scalar habitability score, after which yearly aggregates are computed to analyze temporal trends.

2.1 HABITABILITY INDEX DEFINITION

The habitability index $H \in [0, 1]$ is defined as:

$$H = (S_{size} \times S_{flux} \times S_{orbit} \times S_{star})^{1/4}. \tag{1}$$

Each component ranges between 0 and 1. The four components are multiplied together and the fourth root is taken to preserve the 0-1 scale. This structure ensures that a very low score in any one dimension reduces the overall index. A planet cannot achieve a high habitability score if one critical condition is highly unfavorable.

2.2 PLANETARY SIZE AND COMPOSITION SCORE (S_{size})

The size score is defined using a triangular function of planetary radius R_p:

$$S_{size}(R_p) = \begin{cases} 0, & R_p \leq R_{min} \text{ or } R_p \geq R_{max}, \\ \frac{R_p - R_{min}}{R_{peak} - R_{min}}, & R_{min} < R_p < R_{peak}, \\ \frac{R_{max} - R_p}{R_{max} - R_{peak}}, & R_{peak} \leq R_p < R_{max}. \end{cases} \tag{2}$$

Typical parameter choices are $R_{min} \approx 0.5R_\oplus$, $R_{peak} \approx 1.0R_\oplus$, and $R_{max} \approx 1.8R_\oplus$.

The score increases as planetary radius approaches Earth's size and reaches its maximum near one Earth radius. It decreases gradually as the planet becomes either too small or too large. Planets outside the interval $[R_{min}, R_{max}]$ receive a score of zero. This reflects the idea that very small planets may not retain stable atmospheres, while large planets are more likely to be gas-dominated rather than rocky.

2.3 FLUX / TEMPERATURE SCORE (S_{flux})

The flux score is defined using insolation flux S relative to Earth:

$$S_{flux}(S) = \begin{cases} 0, & S \leq S_{min} \text{ or } S \geq S_{max}, \\ \frac{S - S_{min}}{S_{peak} - S_{min}}, & S_{min} < S < S_{peak}, \\ \frac{S_{max} - S}{S_{max} - S_{peak}}, & S_{peak} \leq S < S_{max}. \end{cases} \tag{3}$$

Typical parameter values are $S_{min} \sim 0.3$, $S_{peak} \sim 1.0$, and $S_{max} \sim 2.0$.

The highest score occurs when stellar energy input is close to Earth's level. As the planet receives either less or significantly more energy, the score decreases. Outside the defined interval, the score becomes zero. This shows the classical habitable zone, where liquid water is most plausible within a moderate energy range.

2.4 ORBITAL STABILITY SCORE (S_{orbit})

Orbital eccentricity e defines the eccentricity score:

$$S_{ecc}(e) = \begin{cases} 1, & 0 \leq e \leq e_0, \\ \max\left(0, 1 - \frac{e - e_0}{e_{max} - e_0}\right), & e_0 < e \leq e_{max}, \\ 0, & e > e_{max}. \end{cases} \tag{4}$$

Typical parameter values are $e_0 \approx 0.1$ and $e_{max} \approx 0.6$.

If a period-based penalty $S_P(P)$ is included, the orbital score is defined as:

$$S_{orbit} = \sqrt{S_{ecc} \cdot S_P}. \tag{5}$$

Planets with nearly circular orbits receive full score. As eccentricity increases, the score decreases linearly. Highly elliptical orbits can produce large seasonal variations in stellar heating, reducing long-term climate stability. If included, the period-based penalty further reduces scores for extremely short-period systems.

2.5 STELLAR ENVIRONMENT SCORE (S_{star})

Stellar age t_* is scored using:

$$S_t(t_*) = \begin{cases} 0, & t_* \leq t_{min} \text{ or } t_* \geq t_{max}, \\ \frac{t_* - t_{min}}{t_{peak} - t_{min}}, & t_{min} < t_* < t_{peak}, \\ \frac{t_{max} - t_*}{t_{max} - t_{peak}}, & t_{peak} \leq t_* < t_{max}. \end{cases} \tag{6}$$

Typical values are $t_{min} \sim 0.5$ Gyr, $t_{peak} \sim 4.5$ Gyr, and $t_{max} \sim 10$ Gyr.

The stellar environment score combines age, effective temperature, and metallicity:

$$S_{star} = (S_t \cdot S_{Teff} \cdot S_{[Fe/H]})^{1/3}. \tag{7}$$

The stellar age score is for the stars that are old enough to be stable but not so old that they have evolved significantly. Effective temperature and metallicity account for stellar type and chemical composition. Combining these factors ensures that the host star contributes meaningfully to overall habitability.

2.6 TIME-SERIES QUANTITIES

For each discovery year y, yearly summaries are defined as:

$$\bar{H}(y) = \frac{1}{N_y} \sum_{i=1}^{N_y} H_i, \tag{8}$$

$$N_{hi}(y) = \#\{i : H_i > H_{thr}, \text{ discovery year } = y\}, \tag{9}$$

$$f_{hi}(y) = \frac{N_{hi}(y)}{N_y}. \tag{10}$$

The yearly mean $\bar{H}(y)$ measures whether average habitability is increasing or decreasing over time. The high-index count $N_{hi}(y)$ and fraction $f_{hi}(y)$ indicate how frequently potentially habitable planets appear among discoveries in each year. Together, these quantities allow assessment of how the statistical profile of discovered planets changes across discovery time.

3 Data and Preprocessing

3.1 Data Source

The dataset used in this study was obtained from the NASA Exoplanet Archive and contains over 200 variables describing planetary, orbital, stellar, and discovery-related characteristics. Because the goal of this project is to construct a composite habitability index, only variables directly connected to the methodological framework were retained. These variables represent the four major dimensions of the index: planetary size, stellar energy input, orbital stability, and stellar environment. In addition, discovery metadata were kept to support the temporal and comparative analyses across years, methods, and facilities.

3.2 Variable Selection and Dimensional Reduction

Although the full dataset contains a large number of variables, many of them are either descriptive, redundant, highly specialized, or not directly relevant to planetary habitability. To reduce dimensionality and keep the analysis interpretable, a smaller subset of variables was selected using domain-based filtering. This step was guided by the habitability framework used in this study, where the final index is based on size, flux, orbital characteristics, and host star properties.

The final variables retained for analysis were:

- `planet_radius_earth_radius`
- `insolation_flux_earth_flux`
- `equilibrium_temperature_k` (retained as a supplementary variable)
- `eccentricity`
- `orbital_period_days`
- `stellar_age_gyr`
- `stellar_effective_temp_k`
- `stellar_metallicity_dex`
- `discovery_year`
- `discovery_method`
- `discovery_facility`

This reduction was necessary for two reasons. First, it removed variables that do not contribute meaningfully to the habitability score. Second, it made the final model easier to interpret, since each retained variable has a direct physical role in the scoring framework.

3.3 Data Cleaning

After selecting the relevant variables, the data were cleaned to prepare them for scoring and analysis. All quantitative variables were converted to numeric format to ensure they could be used in mathematical transformations and piecewise scoring functions. This step was necessary because large public datasets often contain mixed formatting, missing symbols, or imported values that are not automatically stored as numeric values.

Missing values were handled using median imputation for the continuous variables. The median was chosen instead of the mean because it is more robust to skewness and extreme observations, both of which are common in exoplanet data. This approach allowed the analysis to retain a larger number of planets while reducing the influence of outliers on the imputed values. Since the goal of this stage was to construct a stable exploratory habitability index rather than estimate causal relationships, median imputation provided a practical balance between completeness and robustness.

Variables such as discovery method and discovery facility were retained as categorical descriptors and were not imputed numerically, since they were used only for grouped comparisons in later sections.

3.4 Preparation for Habitability Scoring

The cleaned variables were then organized according to the four components of the habitability index:

- **Size component:** planetary radius
- **Flux component:** insolation flux
- **Orbital component:** eccentricity
- **Stellar environment component:** stellar age, stellar effective temperature, and stellar metallicity

This structure follows the methodological framework of the study, where each component is transformed to a score between 0 and 1 and then combined through a geometric mean. Organizing the data in this way ensures that each retained variable contributes directly to a scientifically interpretable dimension of habitability.

3.5 Rationale for Preprocessing Choices

The preprocessing decisions were made to support both interpretability and consistency with the proposed habitability framework. The reduction from more than 200 variables to a smaller set of physically meaningful features ensured that the analysis remained focused on characteristics most relevant to potential habitability. Numeric conversion and missing-value treatment were required to make the data suitable for score construction. Overall, the preprocessing stage established a clean and structured dataset that could be used to calculate the composite habitability index and examine its variation across discovery year, detection method, and facility.

4 Results and Discussion

4.1 Overall Distribution of Habitability

The computed habitability index values are generally low across the dataset. The mean habitability score is approximately 0.0029, with a median value of zero, indicating that most discovered exoplanets do not satisfy multiple favorable conditions simultaneously. Out of 5,788 planets, only 31 exceed the threshold of $H > 0.3$, representing approximately 0.54% of the dataset.

This result suggests that planets with Earth-like characteristics are extremely rare within currently detected exoplanets. This aligns with the expectation that most detected planets are either gas giants or exist in environments with extreme stellar conditions, making them unfavorable for habitability.

4.2 Temporal Trends in Habitability

Figure 1 shows the mean habitability index by discovery year, while Figure 2 shows the fraction of high-habitability planets ($H > 0.3$) over time.

Prior to approximately 2012, the mean habitability index remains effectively zero, reflecting the early focus of detection methods on large, close-in planets such as hot Jupiters. After 2012, a noticeable increase in both mean habitability and the fraction of high-scoring planets is observed.

This shift corresponds to the introduction of more sensitive detection missions, particularly transit-based surveys such as the Kepler mission. These technologies improved the ability to detect smaller, Earth-sized planets in more favorable orbital regions. However, despite this improvement, the overall habitability levels remain low, and fluctuations across years indicate that high-scoring planets are still discovered infrequently and irregularly.

4.3 Comparison Across Detection Methods

Figure 3 presents the mean habitability index by discovery method.

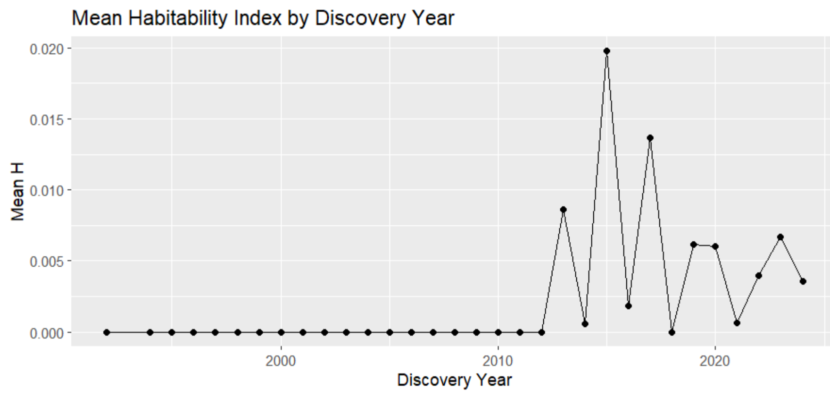

Figure 1: Mean Habitability Index by Discovery Year

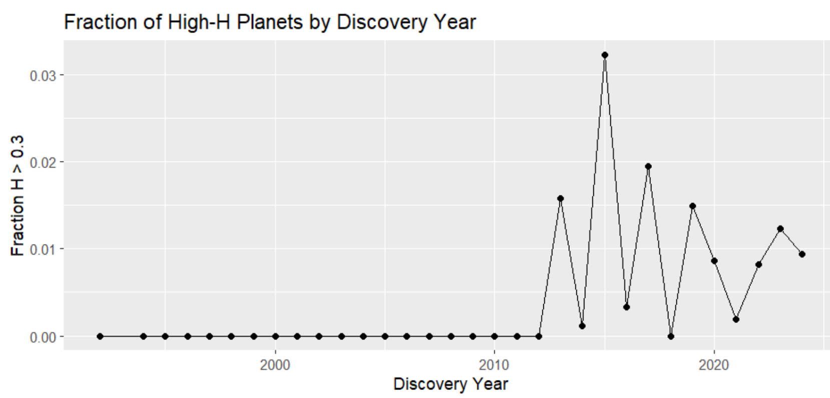

Figure 2: Fraction of High-Habitability Planets by Discovery Year

Among detection methods, Transit Timing Variations show the highest average habitability score. However, this method has a relatively small sample size, and the high mean is influenced by a limited number of observations. In contrast, the Transit and Radial Velocity methods, which account for the majority of detected planets, show lower mean habitability values.

This suggests that widely used detection methods are effective at identifying planets but are not specifically optimized for detecting highly habitable environments. Instead, they tend to detect planets that are easier to observe, which may not necessarily align with Earth-like conditions.

4.4 Comparison Across Discovery Facilities

Figure 4 shows the mean habitability index by discovery facility for facilities with more than 10 detected planets.

Certain facilities, such as the Calar Alto Observatory, show higher mean habitability scores compared to others. However, similar to detection methods, these results may be influenced by smaller sample sizes and specific observational strategies.

Large survey missions, such as the Kepler Space Telescope and TESS, contribute the majority of detected planets but exhibit lower average habitability scores. This reflects the trade-off between large-scale detection capability and the ability to identify highly specific Earth-like conditions.

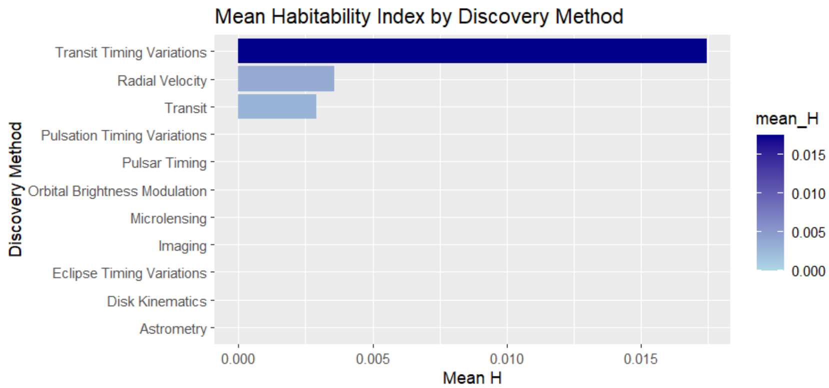

Figure 3: Mean Habitability Index by Discovery Method

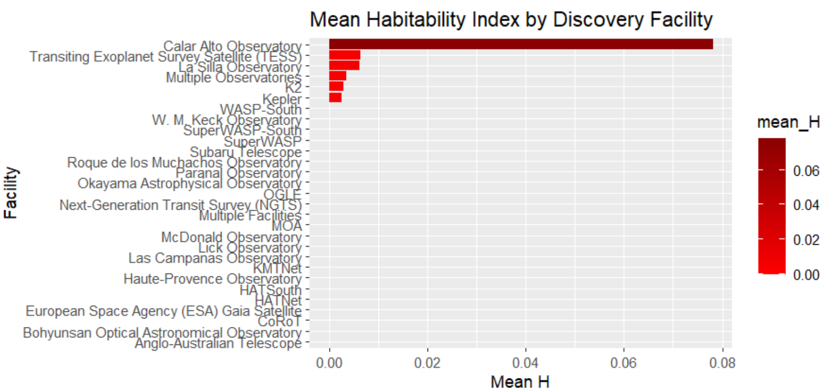

Figure 4: Mean Habitability Index by Discovery Facility

4.5 Interpretation and Implications

The results indicate that while technological advancements have increased the number of detected exoplanets and improved the ability to identify smaller and potentially more Earth-like planets, the discovery of highly habitable planets remains rare.

The increase in mean habitability and high-index fraction after 2012 suggests that modern detection techniques are moving in the right direction. However, the consistently low values of the habitability index highlight the difficulty of simultaneously satisfying all necessary conditions for life.

These findings suggest that future missions may need to focus more specifically on identifying planets within habitable zones and with stable orbital and stellar conditions, rather than primarily maximizing detection volume.

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